Student's Name……...Stream……...

ST JAMES LEGACY SECONDARY SCHOOL-KITAGOBWA

S.2 END OF TERM 1, 2026

Uganda Certificate of Education

MATHEMATICS

TIME: 2 hours 15 minutes

INSTRUCTIONS:

*This paper has one PART; **I .** It has **four** examination items.*

*All items are **compulsory**.*

*Answer **four** examination items in all.*

*Any additional item(s) answered will **not** be scored.*

Graph paper is provided.

Silent, non-programmable scientific calculators and mathematical tables with a list of formula may be used.

PART I

Answer all items from this part

ITEM 1

On their Prom party, five students plan to buy their class teacher a gift. The initial plan was that each one buys their own gift to give to the class teacher but later they realized that If they do it as a group would be cheaper. So they mobilized themselves and each person contributed as follows;

Student A contributed one hundred thirty-three thousand two hundred fifty shillings.

B contributed $\mathbf{1}\frac{1}{2}$ of the amount of A.

C contributed 20% more than that of A.

D contributed a fraction which is $\frac{1}{5}$ less that of B of the amount of A contributed.

E She contributed the balance required to make shs 800,000 which was the cost of the gift. On the day of Prom, the money was given to student E to pick the gift from the nearby shop. From school, she used a boda-boda moving at $10kmh^{-1}$, they moved for 30 minutes due east and then 15 minutes due north to reach the shop.

Tasks:

a). Help to direct another student to locate the shop from school in coordinate form.

b). What amount did each student contributed and who of the five contributed more money.

c). In which other way could they contribute the money more fairly and how much would each contribute that way?

ITEM 2

A certain tourist in U.S.A is planning to visit the following places in Africa next month. The tourist has highlighted the following places on the map of Africa. These places include: - South Africa, Burundi, Kampala, Dodoma, Accra, Bujumbura, Nairobi, Ghana, Johannesburg, Tanzania, Uganda, Kenya, Cairo, Kigali, Egypt, and Rwanda.

Mr. Apeku Elias is an agent for three companies of soda; coca cola, Pepsi, and Riham beverages. He earns 10% commission on every carton of soda sold. He sells each carton of coca cola at ugx.11,000, Pepsi at ugx.11,000 and Riham at ugx.10,000 on a good day, he sells 30 cartons of coca cola, 20 cartons of Pepsi and 10 cartons of Riham.

Task

a) Help the tourist to sort the places by use of an arrow diagram.

b) List the members of the

(i) Domain

(ii) Range.

(iii) What is the relationship between the domain and the range

(iii) Describe the type of mapping in the arrow diagram in (a) above.

c) Help Apeku to know the commission earned on a good day from all the sell

ITEM 3

A bus operator in Jinja and a motorcycle courier in Kampala need to exchange a parcel. At exactly 9:00 a.m., they both start their journeys toward each other along the highway.

- The bus leaves Jinja heading to Kampala at a steady speed of 44km/h
- The motorcycle leaves Kampala heading to Jinja at a constant speed of 25km/h at the same time
- The total distance between Kampala and Jinja is 110km

Hint: (Use Kampala as the starting reference point, where Distance = 0km)

Task:

a) On the same axes draw a distance-time graph for the bus and motorcycle.

b) Use the graph to determine:

i) When and at what distance from Kampala they met

ii) The distance the bus had travelled before meeting the motorcycle

ITEM 4

A delivery drone at a central distribution center (Origin O) needs to deliver a medical supply to medical clinic located at point B. However, the drone must first stop at a charging station locate at point A to pick up a specialized battery

- The charging station (point A) is located at (4,3)
- The medical clinic (point B) is located at (12,9)

The drone's path from charging station to the clinic is represented by displacement vector $\overrightarrow{AB}$

Hints: On a coordinate grid **1 unit** represents **2 kilometers**

The drone requires **0.05 liters** of fuel per kilometer

The cost of fuel is **Ugx 5000 per liter**

Task:

a) i) Sketch the motion of the drone

ii)Represent the position of charging station and the medical clinic from a central distribution center in column vector form

iii) Determine the displacement vector $\overrightarrow{AB}$

b) How long is displacement vector **AB?**

c) How much is needed for fuel for the whole journey from charging station A to medical clinic B?

WINNERS DO NOT QUIT

THE END